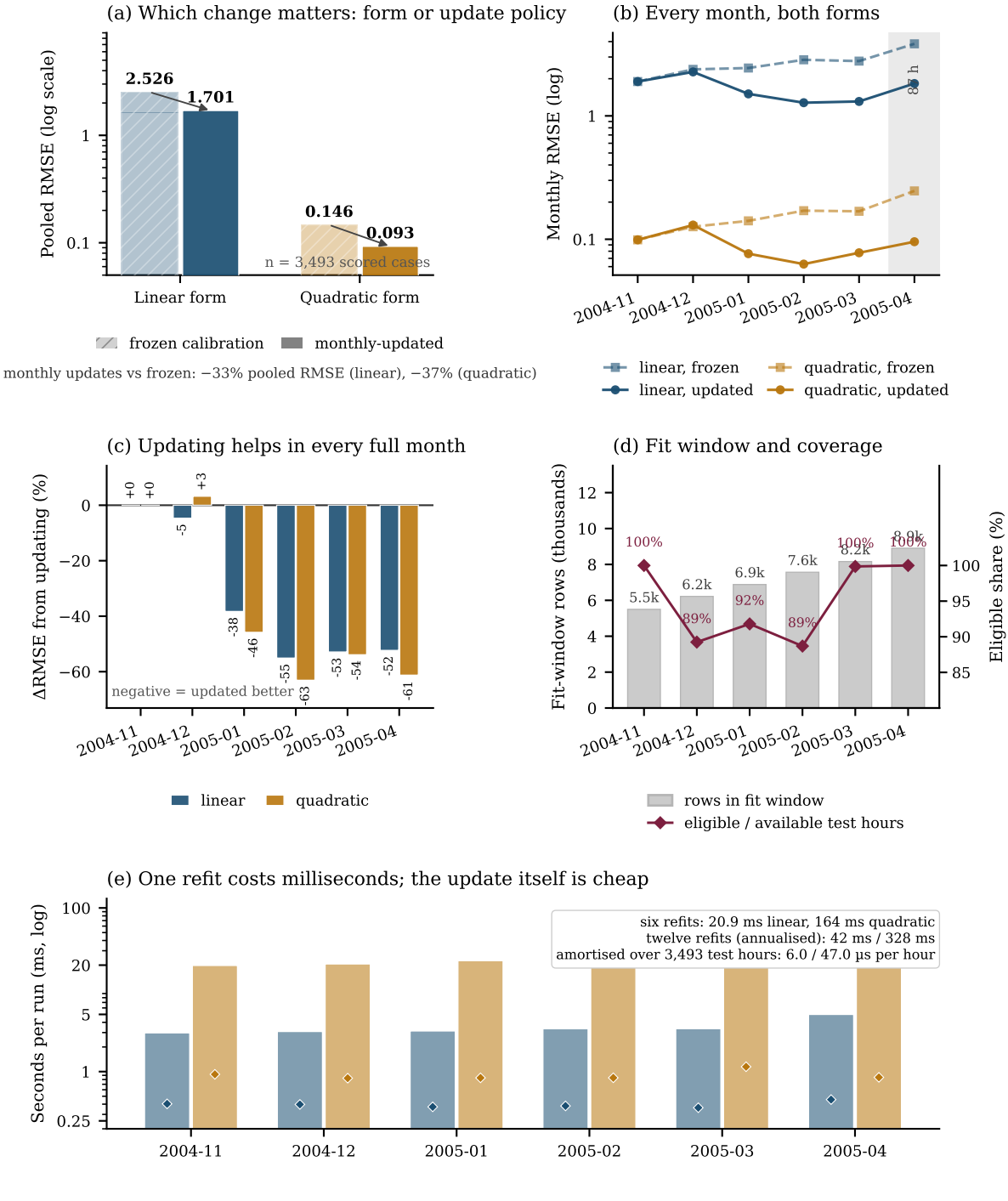
(a) Which change matters: form or update policy
Pooled RMSE (log scale)
2.526
1.701
0.146
0.093
n = 3,493 scored cases
Linear form
Quadratic form
frozen calibration
monthly-updated
monthly updates vs frozen: −33% pooled RMSE (linear), −37% (quadratic)
(b) Every month, both forms
Monthly RMSE (log)
8.7 h
linear, frozen
linear, updated
quadratic, frozen
quadratic, updated
(c) Updating helps in every full month
ΔRMSE from updating (%)
negative = updated better
linear
quadratic
(d) Fit window and coverage
Fit-window rows (thousands)
Eligible share (%)
rows in fit window
eligible / available test hours
(e) One refit costs milliseconds; the update itself is cheap
Seconds per run (ms, log)
six refits: 20.9 ms linear, 164 ms quadratic
twelve refits (annualised): 42 ms / 328 ms
amortised over 3,493 test hours: 6.0 / 47.0 µs per hour
linear: fit (median of repeats)
quadratic: fit (median of repeats)
prediction pass over the month's test cohort